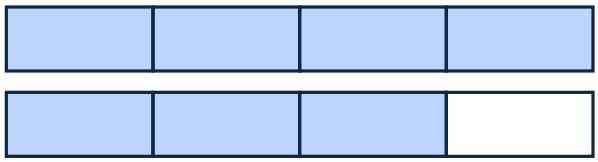
Mixed Numbers

Whole ones and pieces: 1 3/4 means 1 whole and 3/4.

7/4 is 1 whole and 3/4 more



$$\frac{7}{4} = 1\,\frac{3}{4}$$

Say: one and three fourths.

Back to a fraction: count all the pieces

$$2\,\frac{1}{3} = \frac{3}{3} + \frac{3}{3} + \frac{1}{3} = \frac{7}{3}$$

Each whole is 3 thirds: 2 wholes are 6 thirds, plus 1 more is 7 thirds.

Add: wholes with wholes, pieces with pieces

$$2\,\frac{1}{5} + 1\,\frac{3}{5} = 3\,\frac{4}{5}$$

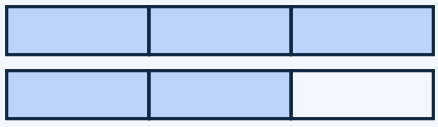
$$1\,\frac{3}{4} + 2\,\frac{2}{4} = 3\,\frac{5}{4} = 4\,\frac{1}{4}$$

5/4 is another whole and 1/4.

Take away the same way

$$3\,\frac{4}{6} - 1\,\frac{1}{6} = 2\,\frac{3}{6}$$

Your turn!



$$\frac{5}{3} = \boxed{1}\,\frac{\boxed{2}}{3}$$

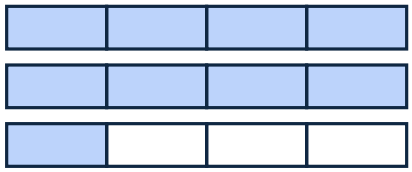
For teachers

Goal: mixed numbers with the same bottom number: rename, add and subtract (4.NF.3c). Say "How many whole bars? How many pieces left over?" Add the wholes, then the pieces.

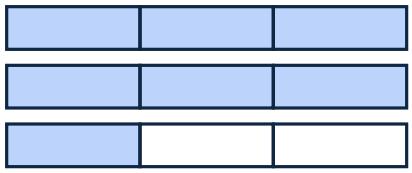
Mixed Numbers: Practice

Whole ones first, then the pieces.

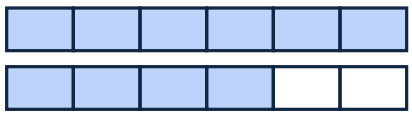
Part 1 Write the mixed number Count the pieces, then the whole bars.

1) 

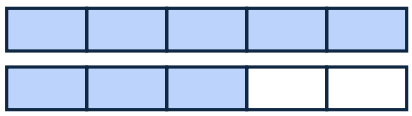
$\frac{9}{4} = 2\frac{1}{4}$

2) 

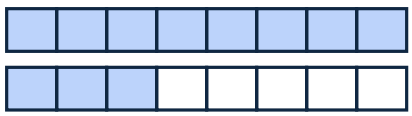
$\frac{7}{3} = 2\frac{1}{3}$

3) 

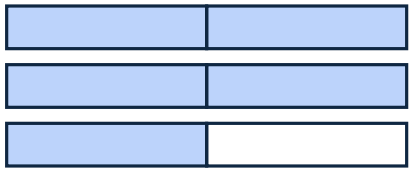
$\frac{10}{6} = 1\frac{4}{6}$

4) 

$\frac{8}{5} = 1\frac{3}{5}$

5) 

$\frac{11}{8} = 1\frac{3}{8}$

6) 

$\frac{5}{2} = 2\frac{1}{2}$

Mixed Numbers: Practice (continued)

Part 2 Fraction to mixed number No help this time.

7) $\frac{9}{2} = 4\frac{1}{2}$

8) $\frac{13}{4} = 3\frac{1}{4}$

9) $\frac{8}{3} = 2\frac{2}{3}$

10) $\frac{12}{5} = 2\frac{2}{5}$

11) $\frac{15}{4} = 3\frac{3}{4}$

12) $\frac{11}{3} = 3\frac{2}{3}$

13) $\frac{17}{6} = 2\frac{5}{6}$

14) $\frac{23}{10} = 2\frac{3}{10}$

15) $\frac{14}{5} = 2\frac{4}{5}$

16) $\frac{19}{8} = 2\frac{3}{8}$

17) $\frac{10}{6} = 1\frac{4}{6}$

18) $\frac{25}{12} = 2\frac{1}{12}$

Mixed Numbers: Practice (continued)

Part 3 Mixed number to fraction No help this time. How many pieces in all?

19) $3\frac{1}{2} = \frac{7}{2}$

20) $2\frac{3}{4} = \frac{11}{4}$

21) $3\frac{1}{3} = \frac{10}{3}$

22) $2\frac{3}{5} = \frac{13}{5}$

23) $4\frac{1}{4} = \frac{17}{4}$

24) $4\frac{2}{3} = \frac{14}{3}$

25) $1\frac{5}{6} = \frac{11}{6}$

26) $2\frac{7}{10} = \frac{27}{10}$

27) $3\frac{2}{5} = \frac{17}{5}$

28) $2\frac{5}{8} = \frac{21}{8}$

29) $3\frac{1}{6} = \frac{19}{6}$

30) $1\frac{1}{12} = \frac{13}{12}$

Mixed Numbers: Practice (continued)

Part 4 Add mixed numbers No help this time. Wholes, then pieces.

31) $2\frac{1}{4} + 1\frac{2}{4} = 3\frac{3}{4}$

32) $1\frac{2}{5} + 1\frac{1}{5} = 2\frac{3}{5}$

33) $1\frac{3}{8} + 1\frac{4}{8} = 2\frac{7}{8}$

34) $2\frac{1}{3} + 1\frac{1}{3} = 3\frac{2}{3}$

35) $2\frac{1}{6} + 2\frac{3}{6} = 4\frac{4}{6}$

36) $2\frac{1}{10} + 3\frac{2}{10} = 5\frac{3}{10}$

37) $1\frac{1}{6} + 1\frac{2}{6} = 2\frac{3}{6}$

38) $2\frac{2}{4} + 1\frac{1}{4} = 3\frac{3}{4}$

39) $1\frac{4}{12} + 1\frac{1}{12} = 2\frac{5}{12}$

Part 5 Make a new whole The pieces make more than 1. Turn them into a whole.

40) $1\frac{3}{4} + 2\frac{3}{4} = 3\frac{6}{4} = 4\frac{2}{4}$

41) $1\frac{3}{5} + 1\frac{4}{5} = 2\frac{7}{5} = 3\frac{2}{5}$

42) $1\frac{5}{8} + 1\frac{6}{8} = 2\frac{11}{8} = 3\frac{3}{8}$

Mixed Numbers: Practice (continued)

Part 5 Make a new whole (continued)

43) $1\frac{2}{3} + 2\frac{2}{3} = 3\frac{4}{3} = 4\frac{1}{3}$

44) $1\frac{5}{6} + 1\frac{3}{6} = 2\frac{8}{6} = 3\frac{2}{6}$

45) $1\frac{7}{10} + 2\frac{9}{10} = 3\frac{16}{10} = 4\frac{6}{10}$

Part 6 Take away mixed numbers No help this time.

46) $3\frac{4}{5} - 1\frac{1}{5} = 2\frac{3}{5}$

47) $3\frac{3}{4} - 1\frac{1}{4} = 2\frac{2}{4}$

48) $3\frac{7}{8} - 1\frac{4}{8} = 2\frac{3}{8}$

49) $3\frac{2}{3} - 1\frac{1}{3} = 2\frac{1}{3}$

50) $3\frac{5}{6} - 1\frac{2}{6} = 2\frac{3}{6}$

51) $3\frac{7}{10} - 1\frac{2}{10} = 2\frac{5}{10}$

52) $4\frac{3}{6} - 2\frac{1}{6} = 2\frac{2}{6}$

53) $4\frac{3}{4} - 1\frac{2}{4} = 3\frac{1}{4}$

54) $3\frac{5}{12} - 1\frac{3}{12} = 2\frac{2}{12}$

Mixed Numbers: Practice (continued)

Part 7 Break a whole first Not enough pieces? Turn one whole into pieces.

55) $3\frac{2}{4} - 1\frac{3}{4} = 2\frac{\boxed{6}}{4} - 1\frac{3}{4} = \boxed{1}\frac{\boxed{3}}{4}$

56) $3\frac{1}{5} - 1\frac{4}{5} = 2\frac{\boxed{6}}{5} - 1\frac{4}{5} = \boxed{1}\frac{\boxed{2}}{5}$

57) $3\frac{1}{8} - 1\frac{7}{8} = 2\frac{\boxed{9}}{8} - 1\frac{7}{8} = \boxed{1}\frac{\boxed{2}}{8}$

58) $3\frac{1}{6} - 1\frac{5}{6} = 2\frac{\boxed{7}}{6} - 1\frac{5}{6} = \boxed{1}\frac{\boxed{2}}{6}$

Part 8 Fraction to mixed number

59) $\frac{22}{6} = \boxed{3}\frac{\boxed{4}}{6}$

60) $\frac{4}{3} = \boxed{1}\frac{\boxed{1}}{3}$

61) $\frac{18}{5} = \boxed{3}\frac{\boxed{3}}{5}$

62) $\frac{16}{5} = \boxed{3}\frac{\boxed{1}}{5}$

63) $\frac{10}{4} = \boxed{2}\frac{\boxed{2}}{4}$

64) $\frac{19}{5} = \boxed{3}\frac{\boxed{4}}{5}$

Mixed Numbers: Practice (continued)

Part 8 Fraction to mixed number (continued)

65) $\frac{8}{6} = 1\frac{2}{6}$

66) $\frac{11}{5} = 2\frac{1}{5}$

67) $\frac{20}{8} = 2\frac{4}{8}$

68) $\frac{36}{10} = 3\frac{6}{10}$

69) $\frac{5}{3} = 1\frac{2}{3}$

70) $\frac{14}{6} = 2\frac{2}{6}$

71) $\frac{8}{5} = 1\frac{3}{5}$

72) $\frac{19}{6} = 3\frac{1}{6}$

73) $\frac{5}{4} = 1\frac{1}{4}$

74) $\frac{22}{10} = 2\frac{2}{10}$

75) $\frac{7}{3} = 2\frac{1}{3}$

76) $\frac{6}{4} = 1\frac{2}{4}$

Mixed Numbers: Practice (continued)

Part 9 Mixed number to fraction

77) $4\frac{2}{5} = \frac{22}{5}$

78) $1\frac{3}{4} = \frac{7}{4}$

79) $4\frac{2}{6} = \frac{26}{6}$

80) $5\frac{2}{5} = \frac{27}{5}$

81) $3\frac{2}{3} = \frac{11}{3}$

82) $4\frac{3}{4} = \frac{19}{4}$

83) $2\frac{6}{8} = \frac{22}{8}$

84) $3\frac{3}{10} = \frac{33}{10}$

85) $5\frac{4}{8} = \frac{44}{8}$

86) $2\frac{1}{5} = \frac{11}{5}$

87) $5\frac{4}{5} = \frac{29}{5}$

88) $1\frac{4}{5} = \frac{9}{5}$

Mixed Numbers: Practice (continued)

Part 9 Mixed number to fraction (continued)

89) $4\frac{1}{3} = \frac{13}{3}$

90) $4\frac{1}{5} = \frac{21}{5}$

91) $2\frac{4}{8} = \frac{20}{8}$

92) $4\frac{2}{4} = \frac{18}{4}$

93) $4\frac{5}{8} = \frac{37}{8}$

94) $4\frac{3}{5} = \frac{23}{5}$

Part 10 Remember? Problems from earlier packets.

95) $\frac{5}{8} + \frac{2}{8} = \frac{7}{8}$

96) $\frac{9}{10} - \frac{4}{10} = \frac{5}{10}$

97) $\frac{7}{12} > \frac{1}{2}$

98) $72 \times 21 = 1{,}512$

99) $2{,}448 \div 4 = 612$

100) Round 6,550 to the nearest hundred: 6,600

Mixed Numbers: Practice (continued)

Part 11 Word problems Draw the bars.

101) Mia ran 1 2/4 miles on Monday and 2 1/4 miles on Tuesday. How far did she run?

Number sentence 1 2/4 + 2 1/4 = 3 3/4

Answer 3 3/4 miles

Draw or show your work

102) A pizza has 8 slices. Ben's family ate 11 slices from two pizzas. How many pizzas is that?

Number sentence 11/8 = 1 3/8

Answer 1 3/8 pizzas

Draw or show your work

103) A ribbon is 4 5/6 feet long. Leo cuts off 2 1/6 feet. How long is the rest?

Number sentence 4 5/6 − 2 1/6 = 2 4/6

Answer 2 4/6 feet

Draw or show your work

104) Kim baked 2 3/4 pans of bread and then 1 3/4 pans more. How many pans in all?

Number sentence 2 3/4 + 1 3/4 = 3 6/4 = 4 2/4

Answer 4 2/4 pans

Draw or show your work

105) Challenge: A jug holds 3 1/4 liters. Sam pours out 1 3/4 liters. How much is left in the jug?

Number sentence 3 1/4 − 1 3/4 = 1 2/4

Answer 1 2/4 liters

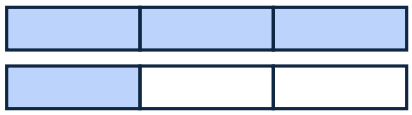
Draw or show your work

Check-In: Mixed Numbers

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Write the mixed number

1) 

$\frac{4}{3} = 1\frac{1}{3}$

Part 2 Fraction to mixed number

2) $\frac{9}{4} = 2\frac{1}{4}$

3) $\frac{11}{6} = 1\frac{5}{6}$

4) $\frac{28}{10} = 2\frac{8}{10}$

Part 3 Mixed number to fraction

5) $3\frac{4}{10} = \frac{34}{10}$

6) $1\frac{5}{10} = \frac{15}{10}$

7) $3\frac{11}{12} = \frac{47}{12}$

Part 4 Add and take away

8) $2\frac{1}{6} + 1\frac{2}{6} = 3\frac{3}{6}$

9) $3\frac{4}{5} - 1\frac{2}{5} = 2\frac{2}{5}$